

NT-C-024 | AI & Machine Learning | Course Knowledge Profile

Identity, purpose, prerequisites, scope and boundaries

Category	Data & AI
Target learners	Learners seeking entry-level applied machine-learning roles.
Prerequisites	Python, basic statistics and algebra foundations.
Approved duration	20-24 weeks / 220-280 guided hours.
Final outcome	Prepare data, train and evaluate machine-learning models, and deliver a documented responsible AI solution.
Assessment profile	Data/AI

Approved tools / platforms

Python, Jupyter, NumPy, pandas, scikit-learn, visualization libraries, Git; optional deep-learning framework.

Knowledge boundaries

Include	Quality rule	Exclude / escalate
The eight approved modules, four sections, named tools, projects and mapped entry-level roles.	Require data provenance, cleaning decisions, train/test separation, suitable metrics, reproducibility and responsible interpretation.	Reject data leakage, invented metrics, unsupported causal claims and hidden manual corrections.
Current product/platform procedures only when version and date are known.	Use primary documentation and record the source version.	Do not invent current commands, prices, tax rates, policies or certification status.
Evidence-based career guidance.	Cite tests, projects, capstone, viva and verified resume evidence.	No guaranteed job, placement, salary, interview or employer claim.

LLM answer pattern: identify the course and version, answer from approved records, cite record IDs, distinguish verified facts from guidance, and state any missing evidence.

NT-C-024 | AI & Machine Learning | NT-C-024-S01 Deep Knowledge

Modules 1-2 | objectives, concepts, evidence and remediation

Section competency	NumPy, pandas and notebooks; Probability, vectors, distributions and metrics
Completion evidence	Data exercises; Concept workbook
Section weight	20% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 1 - Python and data

Objective: NumPy, pandas and notebooks

Observable evidence: Data exercises

Concept	Approved knowledge statement	Application behaviour	Evidence
NumPy	A defined competency within AI & Machine Learning. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when numpy is appropriate; apply it in a AI & Machine Learning task; verify the result.	Data exercises
pandas	A defined competency within AI & Machine Learning. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when pandas is appropriate; apply it in a AI & Machine Learning task; verify the result.	Data exercises
notebooks	A defined competency within AI & Machine Learning. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when notebooks is appropriate; apply it in a AI & Machine Learning task; verify the result.	Data exercises

Module 2 - Math and statistics

Objective: Probability, vectors, distributions and metrics

Observable evidence: Concept workbook

Concept	Approved knowledge statement	Application behaviour	Evidence
Probability	A defined competency within AI & Machine Learning. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when probability is appropriate; apply it in a AI & Machine Learning task; verify the result.	Concept workbook
vectors	A defined competency within AI & Machine Learning. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when vectors is appropriate; apply it in a AI & Machine Learning task; verify the result.	Concept workbook
distributions	A defined competency within AI & Machine Learning. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when distributions is appropriate; apply it in a AI & Machine Learning task; verify the result.	Concept workbook
metrics	A defined competency within AI & Machine Learning. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when metrics is appropriate; apply it in a AI & Machine Learning task; verify the result.	Concept workbook

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Data exercises; Concept workbook, return to Modules 1-2, complete one guided practice and then use a new equivalent test form.

NT-C-024 | AI & Machine Learning | NT-C-024-S02 Deep Knowledge

Modules 3-4 | objectives, concepts, evidence and remediation

Section competency	Cleaning, encoding, scaling and splitting; Regression, classification and baselines
Completion evidence	Prepared pipeline; Compared models
Section weight	25% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 3 - Preprocessing

Objective: Cleaning, encoding, scaling and splitting

Observable evidence: Prepared pipeline

Concept	Approved knowledge statement	Application behaviour	Evidence
Cleaning	A defined competency within AI & Machine Learning. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when cleaning is appropriate; apply it in a AI & Machine Learning task; verify the result.	Prepared pipeline
encoding	A defined competency within AI & Machine Learning. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when encoding is appropriate; apply it in a AI & Machine Learning task; verify the result.	Prepared pipeline
scaling	A defined competency within AI & Machine Learning. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when scaling is appropriate; apply it in a AI & Machine Learning task; verify the result.	Prepared pipeline
splitting	A defined competency within AI & Machine Learning. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when splitting is appropriate; apply it in a AI & Machine Learning task; verify the result.	Prepared pipeline

Module 4 - Supervised learning

Objective: Regression, classification and baselines

Observable evidence: Compared models

Concept	Approved knowledge statement	Application behaviour	Evidence
Regression	Supervised modelling of a continuous outcome. In AI & Machine Learning, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when regression is appropriate; apply it in a AI & Machine Learning task; verify the result.	Compared models
classification	Supervised modelling of a categorical outcome. In AI & Machine Learning, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when classification is appropriate; apply it in a AI & Machine Learning task; verify the result.	Compared models
baselines	A defined competency within AI & Machine Learning. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when baselines is appropriate; apply it in a AI & Machine Learning task; verify the result.	Compared models

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Prepared pipeline; Compared models, return to Modules 3-4, complete one guided practice and then use a new equivalent test form.

NT-C-024 | AI & Machine Learning | NT-C-024-S03 Deep Knowledge

Modules 5-6 | objectives, concepts, evidence and remediation

Section competency	Clustering and dimensionality reduction; Features, validation, tuning and imbalance
Completion evidence	Segmentation study; Validated model
Section weight	25% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 5 - Unsupervised learning

Objective: Clustering and dimensionality reduction

Observable evidence: Segmentation study

Concept	Approved knowledge statement	Application behaviour	Evidence
Clustering	Unsupervised grouping based on similarity. In AI & Machine Learning, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when clustering is appropriate; apply it in a AI & Machine Learning task; verify the result.	Segmentation study
dimensionality reduction	A defined competency within AI & Machine Learning. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when dimensionality reduction is appropriate; apply it in a AI & Machine Learning task; verify the result.	Segmentation study

Module 6 - Model improvement

Objective: Features, validation, tuning and imbalance

Observable evidence: Validated model

Concept	Approved knowledge statement	Application behaviour	Evidence
Features	A defined competency within AI & Machine Learning. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when features is appropriate; apply it in a AI & Machine Learning task; verify the result.	Validated model
validation	Checking input or output against defined rules before trusting it. In AI & Machine Learning, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when validation is appropriate; apply it in a AI & Machine Learning task; verify the result.	Validated model
tuning	A defined competency within AI & Machine Learning. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when tuning is appropriate; apply it in a AI & Machine Learning task; verify the result.	Validated model
imbalance	A defined competency within AI & Machine Learning. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when imbalance is appropriate; apply it in a AI & Machine Learning task; verify the result.	Validated model

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Segmentation study; Validated model, return to Modules 5-6, complete one guided practice and then use a new equivalent test form.

NT-C-024 | AI & Machine Learning | NT-C-024-S04 Deep Knowledge

Modules 7-8 | objectives, concepts, evidence and remediation

Section competency	NLP and computer-vision foundations; Pipelines, deployment basics, monitoring and responsible use
Completion evidence	Applied mini project; Capstone solution
Section weight	30% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 7 - Applied AI

Objective: NLP and computer-vision foundations

Observable evidence: Applied mini project

Concept	Approved knowledge statement	Application behaviour	Evidence
NLP	A defined competency within AI & Machine Learning. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when nlp is appropriate; apply it in a AI & Machine Learning task; verify the result.	Applied mini project
computer-vision foundations	A defined competency within AI & Machine Learning. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when computer-vision foundations is appropriate; apply it in a AI & Machine Learning task; verify the result.	Applied mini project

Module 8 - Delivery and ethics

Objective: Pipelines, deployment basics, monitoring and responsible use

Observable evidence: Capstone solution

Concept	Approved knowledge statement	Application behaviour	Evidence
Pipelines	A defined competency within AI & Machine Learning. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when pipelines is appropriate; apply it in a AI & Machine Learning task; verify the result.	Capstone solution
deployment basics	A defined competency within AI & Machine Learning. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when deployment basics is appropriate; apply it in a AI & Machine Learning task; verify the result.	Capstone solution
monitoring	Collection and interpretation of metrics, logs and events to understand system health. In AI & Machine Learning, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when monitoring is appropriate; apply it in a AI & Machine Learning task; verify the result.	Capstone solution
responsible use	A defined competency within AI & Machine Learning. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when responsible use is appropriate; apply it in a AI & Machine Learning task; verify the result.	Capstone solution

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Applied mini project; Capstone solution, return to Modules 7-8, complete one guided practice and then use a new equivalent test form.

NT-C-024 | AI & Machine Learning | Skill Ontology & Dependency Map

Atomic skills, learning order and evidence strength

Skill ID	Competency	Depends on	Evidence	Type
NT-C-024-SK01	Python and data	Prerequisite foundation	Data exercises	Critical
NT-C-024-SK02	Math and statistics	M01 plus earlier foundations	Concept workbook	Critical
NT-C-024-SK03	Preprocessing	M02 plus earlier foundations	Prepared pipeline	Supporting
NT-C-024-SK04	Supervised learning	M03 plus earlier foundations	Compared models	Supporting
NT-C-024-SK05	Unsupervised learning	M04 plus earlier foundations	Segmentation study	Supporting
NT-C-024-SK06	Model improvement	M05 plus earlier foundations	Validated model	Critical
NT-C-024-SK07	Applied AI	M06 plus earlier foundations	Applied mini project	Supporting
NT-C-024-SK08	Delivery and ethics	M07 plus earlier foundations	Capstone solution	Critical

Evidence-strength hierarchy

Strength	Evidence source	Use
High	Trainer-observed practical, viva, working source/project files and reproducible output.	May support verified mastery when rubric threshold is met.
Medium	Scored mock tests, reviewed assignments and documented portfolio artifacts.	Supports mastery with corroboration.
Low	Resume claim, self-report, attendance or video completion without demonstration.	May trigger verification; cannot alone support Apply Now.

CareerTours must not treat attendance, time spent or content opened as skill mastery.

NT-C-024 | AI & Machine Learning | Assessment Knowledge & Sample Items

Non-live examples for LLM training and evaluation

Sample ID	Type	Question	Approved answer / rubric anchor
NT-C-024-S01-EX01	Evidence/application	What evidence best demonstrates the combined competency of Python and data and Math and statistics?	The learner should provide Data exercises; Concept workbook and explain how the work applies NumPy, pandas and notebooks; Probability, vectors, distributions and metrics. Recall alone is insufficient.
NT-C-024-S01-EX02	Scenario/remediation	A learner reports completing Math and statistics but cannot produce Concept workbook. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-024-S01, request Concept workbook, and allow a new equivalent test only after remediation.
NT-C-024-S02-EX01	Evidence/application	What evidence best demonstrates the combined competency of Preprocessing and Supervised learning?	The learner should provide Prepared pipeline; Compared models and explain how the work applies Cleaning, encoding, scaling and splitting; Regression, classification and baselines. Recall alone is insufficient.
NT-C-024-S02-EX02	Scenario/remediation	A learner reports completing Supervised learning but cannot produce Compared models. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-024-S02, request Compared models, and allow a new equivalent test only after remediation.
NT-C-024-S03-EX01	Evidence/application	What evidence best demonstrates the combined competency of Unsupervised learning and Model improvement?	The learner should provide Segmentation study; Validated model and explain how the work applies Clustering and dimensionality reduction; Features, validation, tuning and imbalance. Recall alone is insufficient.
NT-C-024-S03-EX02	Scenario/remediation	A learner reports completing Model improvement but cannot produce Validated model. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-024-S03, request Validated model, and allow a new equivalent test only after remediation.
NT-C-024-S04-EX01	Evidence/application	What evidence best demonstrates the combined competency of Applied AI and Delivery and ethics?	The learner should provide Applied mini project; Capstone solution and explain how the work applies NLP and computer-vision foundations; Pipelines, deployment basics, monitoring and responsible use. Recall alone is insufficient.
NT-C-024-S04-EX02	Scenario/remediation	A learner reports completing Delivery and ethics but cannot produce Capstone solution. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-024-S04, request Capstone solution, and allow a new equivalent test only after remediation.

Live bank rule: Generate at least three times the live quantity for each section using the approved Data/AI blueprint. Human-review every scored item before initial production release.

NT-C-024 | AI & Machine Learning | Labs, Projects & Scoring Rubrics

Practical proof of learning and ownership

Level	Approved project	Skills evidenced	Required deliverables
Mini 1	Prediction baseline	Preprocessing and supervised ML	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Mini 2	Segmentation study	Clustering and interpretation	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Capstone	Applied ML product	Evaluation, ethics and delivery	Brief; working/source file or reproducible environment; output; validation notes; reflection.

Standard project rubric

Dimension	Weight	What earns credit
Problem framing	15%	Clear objective, user/business need, assumptions and success criteria.
Technical/design accuracy	25%	Correct methods, appropriate tools and sound decisions.
Completeness	15%	Required features, assets, data or workflow are present.
Testing and validation	15%	Important normal, edge, quality or failure cases are checked.
Evidence and reproducibility	15%	Source/working files, setup, inputs, outputs and version information are available.
Documentation	5%	Concise structure, instructions and limitations.
Viva / ownership	10%	Learner can explain decisions, demonstrate work and respond to a change request.

Require data provenance, cleaning decisions, train/test separation, suitable metrics, reproducibility and responsible interpretation.

NT-C-024 | AI & Machine Learning | Career Roles, Evidence & Gap Actions

Explainable top-three occupation matching

Rank	Occupation	Default weighted skills	Minimum evidence	Gap action
1	ML Engineer Trainee	Python 34% [critical]; pipelines 33% [critical]; evaluation 33%	Reproducible model	If a critical skill is below 60 or reproducible model is missing, return Improve First or Needs Evidence with the linked module/project.
2	AI/ML Analyst - Junior	Data 34% [critical]; modelling 33% [critical]; interpretation 33%	Analysis report	If a critical skill is below 60 or analysis report is missing, return Improve First or Needs Evidence with the linked module/project.
3	Data Science Associate	EDA 34% [critical]; statistics 33% [critical]; ML 33%	Portfolio capstone	If a critical skill is below 60 or portfolio capstone is missing, return Improve First or Needs Evidence with the linked module/project.

Course readiness	45% section mocks + 20% mini projects + 25% capstone + 10% viva.
Role readiness	45% role-relevant mastery + 25% projects/capstone + 20% aligned mocks + 10% verified resume evidence.
Evidence confidence	Show separately; Apply Now requires at least 70.
Resume extraction	Use only relevant skills, projects, education, experience and portfolio evidence; minimise personal data before external model processing.

A student may be Apply Now for one mapped occupation and Improve First for another. Return independent evidence and gaps for each role.

NT-C-024 | AI & Machine Learning | FAQs, Misconceptions & LLM Response Pack

Course-specific retrieval and response patterns

Approved FAQ	Answer
Who is AI & Machine Learning for?	Learners seeking entry-level applied machine-learning roles.
What should I know before starting?	Python, basic statistics and algebra foundations.
How long is the approved learning journey?	20-24 weeks / 220-280 guided hours.
What will I be able to demonstrate?	Prepare data, train and evaluate machine-learning models, and deliver a documented responsible AI solution.
Does completion guarantee a job?	No. Completion and role readiness are evidence-based guidance; employment, placement, salary and employer selection are not guaranteed.
What if I fail a section?	CareerTours identifies the failed skill tags and module, assigns targeted practice, and permits an equivalent retest after remediation.

Misconception	Approved correction
Attendance proves mastery	False - require tests and observable practical/project evidence.
One total score proves every role fit	False - calculate each occupation using its weighted skills and evidence.
The LLM can add missing details	False - retrieve approved records or state that evidence is insufficient.
Course completion guarantees employment	False - never claim guaranteed job, placement, salary or interview.
AI & Machine Learning knowledge never changes	False - version-sensitive tools, laws, taxes, security and platform procedures require review.

Retrieval metadata

```
course_code=NT-C-024; corpus_version=3.0;
record_types=course_profile,module,section,skill,assessment_sample,project,occupation_mapping,faq,misconception;
approval_status=approved; effective_date=2026-08-12
```